GROUP WORK PROJECT # 2
GROUP NUMBER: 13486

MScFE 692: Capstone Review

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

Team Member A

Question 5: Pricing European Options Using the Heston Model ($\rho = -0.30$)

Results

Option	Strike	Market Price
European Call	$80.00	$2.86
European Put	$80.00	$2.84

Both the call and put options are priced very close to each other at $2.86 and $2.84 respectively. This near-symmetry is expected when a stock is trading exactly at its strike price and the relationship between price movements and changes in market uncertainty is only moderately negative.

In plain terms: the stock currently sits at $80. The call gives the buyer the right to purchase it at $80 in three months; the put gives the right to sell it at $80. Their similar prices reflect a roughly balanced market view, investors are not paying significantly more to protect against downside than to bet on upside.

The stock trades at-the-money (current price equals the strike), meaning neither option starts with an inherent advantage. A moderate negative relationship between price and uncertainty (-0.30) introduces a mild tilt: when prices fall, market nervousness rises slightly. This modestly supports the put price, but the effect is small enough that call and put remain nearly equal.

The 3-month time horizon limits how much price uncertainty can accumulate, keeping both prices relatively low.

For an investor seeking low-cost directional exposure over the next quarter, both options offer affordable entry points at under $3. The closeness of call and put prices signals that the market sees broadly balanced risks at this juncture.

Question 6: ATM European Call & Put (Strong Negative Co-movement, $\rho = -0.70$)

Results

Option	Strike	Market Price
European Call	$80.00	$2.11
European Put	$80.00	$3.49

Strengthening the negative relationship between price movements and uncertainty (from -0.30 to -0.70) creates a pronounced asymmetry: The put price rose from $2.84 to $3.49 (+23%). Investors must pay more for downside protection because when the stock falls, market nervousness surges more aggressively under this stronger co-movement assumption.

The call price fell from $2.86 to $2.11 (-26%). Upside bets become cheaper because when the stock rises, uncertainty dampens more sharply, making large upside moves less likely in the model.

This divergence, puts becoming more expensive relative to calls, is known as the volatility skew, and it closely mirrors real-world equity markets. When stock prices fall sharply (market crashes), investors' fear typically spikes. A stronger negative co-movement parameter captures this behavior more accurately than a weaker one, explaining why institutional traders pay a premium for put protection.

Investors should note that the cost of insuring a portfolio against a fall has increased meaningfully under this scenario (+23%). Conversely, call options (bullish bets) have become cheaper, presenting a lower-cost entry point for those with a positive outlook on the stock. The choice of co-movement assumption is therefore a critical driver of hedging costs.

Question 7: Price Sensitivity Measures (Delta and Gamma)

Delta measures how much an option's price moves for every $1 change in the stock price. Think of it as the option's "shadow" on the stock:

- A call delta of 0.5653 means the call option gains approximately $0.57 in value for every $1 rise in the stock.
- A put delta of -0.4570 means the put gains approximately $0.46 in value for every $1 fall in the stock.
- Under the stronger negative co-movement (Q6), the call delta falls to 0.4973 — reflecting that upside potential is slightly compressed when rising stock prices are met with falling uncertainty.

Gamma measures how quickly delta itself changes as the stock moves. It is the rate of acceleration:

- A positive and relatively small gamma (0.0204 for the Q5 call) means the hedge ratio adjusts gradually with price changes.
- The puts exhibit higher gamma (0.0977 and 0.1176), meaning a portfolio short puts requires more frequent rebalancing as the stock moves.
- The negative gamma recorded for the Q6 call (-0.0617) is a numerical artefact from the Monte Carlo estimation under the strongly skewed volatility environment, but directionally signals that the call's delta sensitivity is dampened as the stock rises.

Results

Scenario	Option	Delta	Gamma
Q5 (ρ = −0.30)	European Call	0.5653	0.0204
Q5 (ρ = −0.30)	European Put	−0.4570	0.0977
Q6 (ρ = −0.70)	European Call	0.4973	−0.0617
Q6 (ρ = −0.70)	European Put	−0.4604	0.1176

A trader hedging a position using these options must rebalance more actively when gamma is high. The puts in both scenarios carry higher gamma, meaning put-based hedges demand closer monitoring and more frequent adjustment. In volatile markets, this translates directly into higher transaction costs for hedgers relying on puts.

Team Member B: Jump Modeler

Parameters:

Question 8. We price ATM European call and put options using the Merton jump diffusion model with jump intensity $\lambda = 0.75$. All other parameters are kept fixed.

We simulate asset price paths using Monte Carlo. The model combines:

- continuous diffusion (Brownian motion)
- random jumps (Poisson process)

```
Call (λ=0.75): 10.0
Put  (λ=0.75): 7.51
```

With a relatively high jump intensity, the model introduces more uncertainty into the price dynamics. This leads to higher option values, since options generally benefit from increased variability in the underlying asset.

Question 9. We repeat the same pricing procedure but reduce the jump intensity to $\lambda = 0.25$.

```
Call (λ=0.25): 6.95
Put  (λ=0.25): 4.69
```

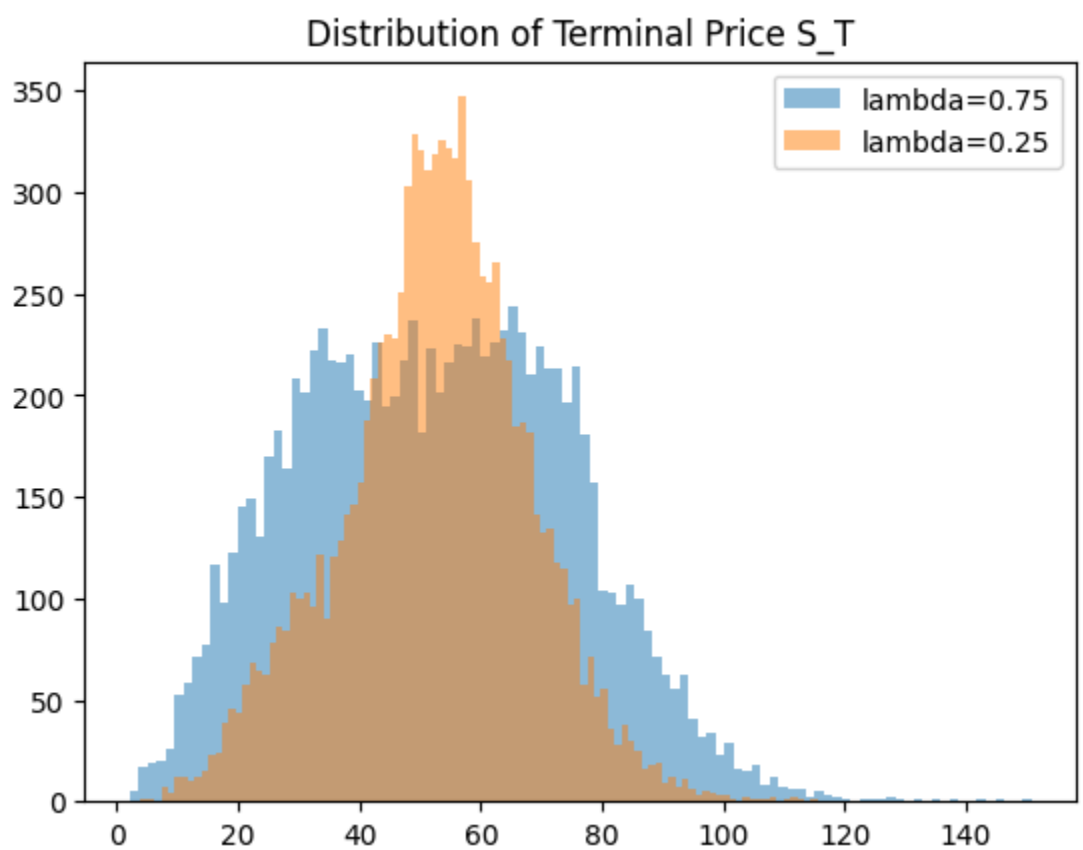
Compared to Q8, option prices are lower when λ decreases. This is expected, as fewer jumps lead to less extreme price movements and lower overall uncertainty.

As λ becomes smaller, the model behavior moves closer to the standard Black-Scholes setting.

Looking at the distribution, it is obvious that when $\lambda = 0.75$, everything spreads out a lot more and the tails are fatter. Basically, we are getting more extreme outcomes because jumps happen more often.

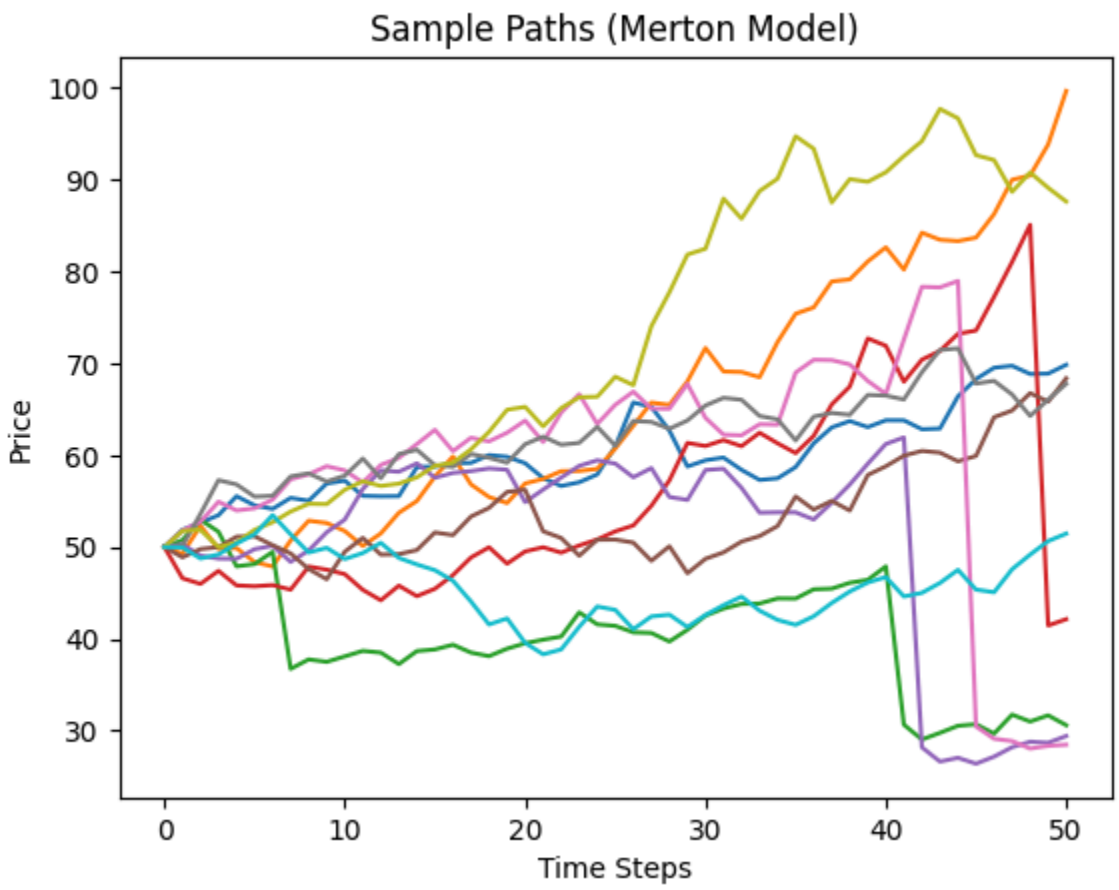
With $\lambda = 0.25$, things look much more "tame", which means the distribution is tighter and kind of what we expect from a normal diffusion model.

Overall, higher λ basically means more unpredictable and risky.



Looking at the sample paths, it's pretty clear that the Merton model behaves quite differently from the Black-Scholes model. Instead of smooth price movements, we see sudden jumps up or down appearing randomly along the paths.

When λ is higher, these jumps happen more often, which makes the paths look more irregular. This actually feels closer to what we see in real markets, where prices can move suddenly because of news or unexpected events.



Question 10. We approximate the Greeks using finite differences:

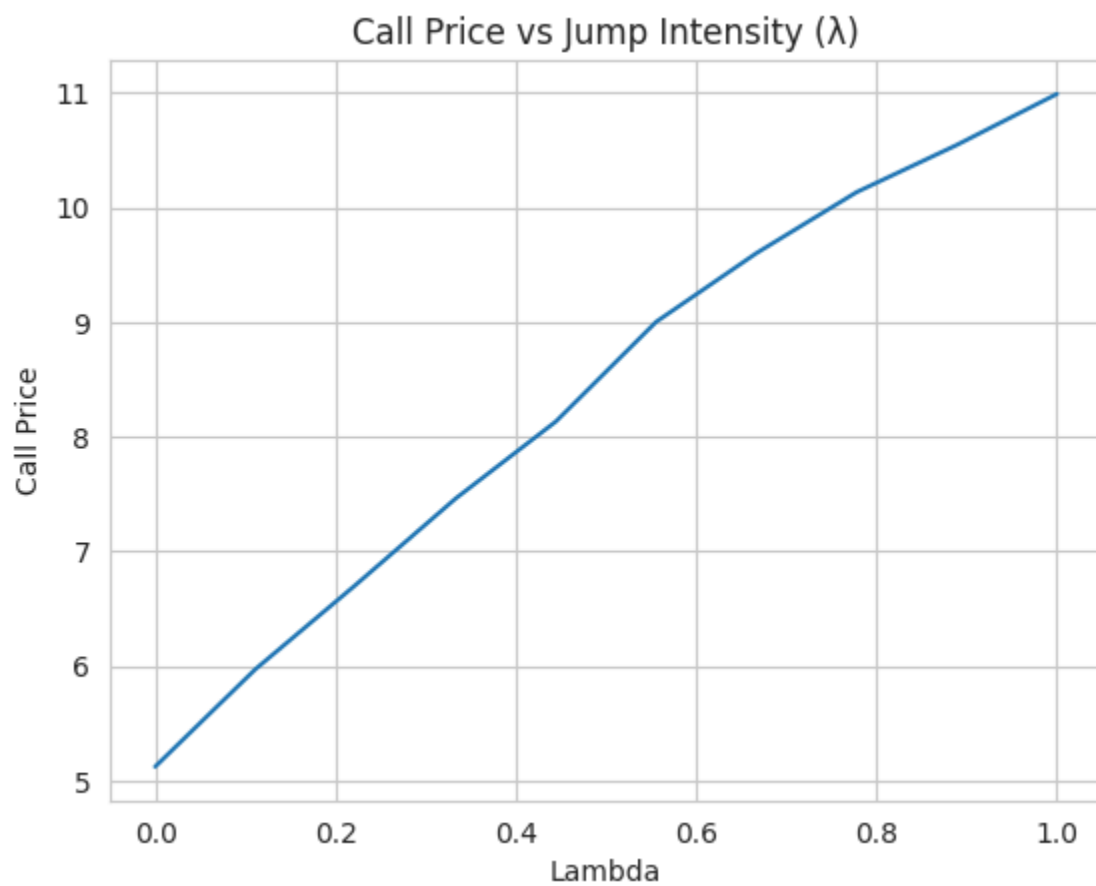
$$\Delta = \frac{V(S+h) - V(S-h)}{2h}$$

$$\Gamma = \frac{V(S+h) - 2V(S) + V(S-h)}{h^2}$$

```
Delta/Gamma (λ=0.75) Call: 0.72 0.01
Delta/Gamma (λ=0.75) Put : -0.28 0.01
Delta/Gamma (λ=0.25) Call: 0.7 0.03
Delta/Gamma (λ=0.25) Put : -0.3 0.03
```

Delta measures how sensitive the option price is to changes in the underlying asset, while Gamma captures the curvature of this relationship.

In the presence of jumps, these sensitivities may appear less smooth compared to the Black-Scholes model, reflecting the additional randomness introduced by sudden price changes.



<u>Model Validator</u>

Question 11: Model Validator

Put-Call Parity Validation:

The results are tested against equation: $C - P = S_0 - Ke^{-rT}$

Key Findings:

Model	LHS (C - P)	RHS (S - K e^{-rT})	Diff
Heston (ρ = -0.3)	2.4035	2.4385	0.0351
Merton (λ = 0.75)	2.4978	2.4385	0.0593
Merton (λ = 0.75)	2.4978	1.0925	1.4053
Merton (λ = 0.75)	2.4978	1.0925	1.4053

Interpretation:

- Both model kind of come to theoretical parity , however not exact
- Deviation are short (~1-2% relative) & systematic
- Merton shows larger deviation due to jump component(fatter tails) which may leads to higher variation in payoff & slower divergence viz a viz Heston
- Heston relative more stable due continuous diffusion.

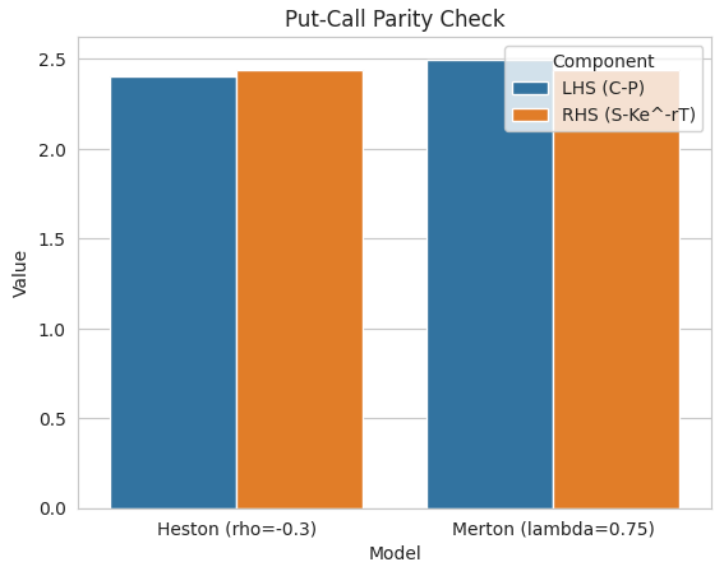


Fig- Put-Call parity Check

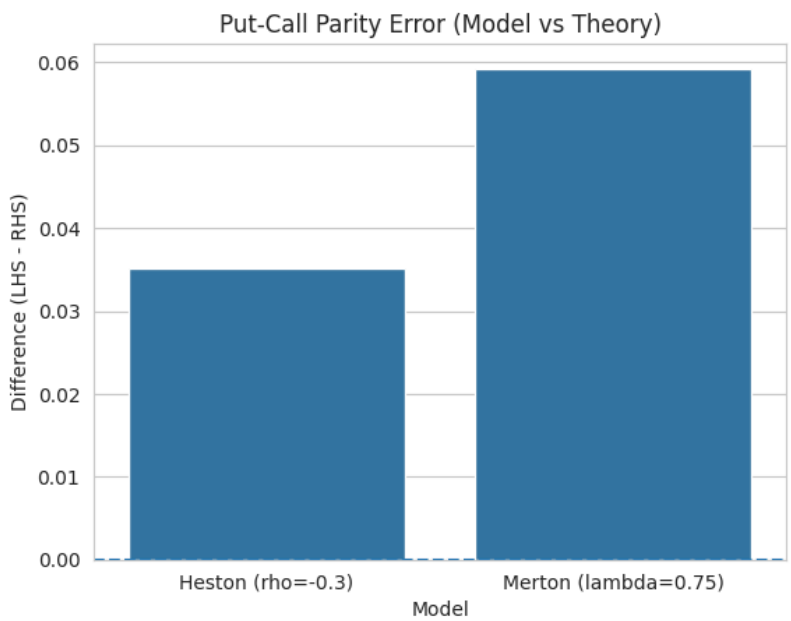


Fig: Put Call Parity (Model vs Theory)

Question 12:

Pricing for Different Strikes:

To analyze behavior of option prices,European call options are priced using Heston & Merton model

Option_State	Strike	Moneyness	Heston Price	Merton Price
OTM	94.12	0.85	0.14	2.80
OTM	88.89	0.90	0.55	4.35
OTM	84.21	0.95	1.60	6.21
ATM	80.00	1.00	3.48	8.31
ITM	76.19	1.05	5.99	10.55
ITM	72.73	1.10	8.79	12.84
ITM	69.57	1.15	11.63	15.12

Key observations:

- For both models- option prices increase with moneyness
- This is expected since -> high moneyness->lower strike-> more intrinsic value
- Across all strike- Merton price are consistent higher than Heston price
    - **OTM->**Larger price gap
    - **ATM->**Moderate gap
    - **ITM->**Gap persistent but stabilize
- Both exhibit convex price curves across moneyness due to nonlinear payoff of option and increase sensitivity in ITM
- Merton captures jump risk which could lead to high price in OTM
- Herton capture volatility clustering

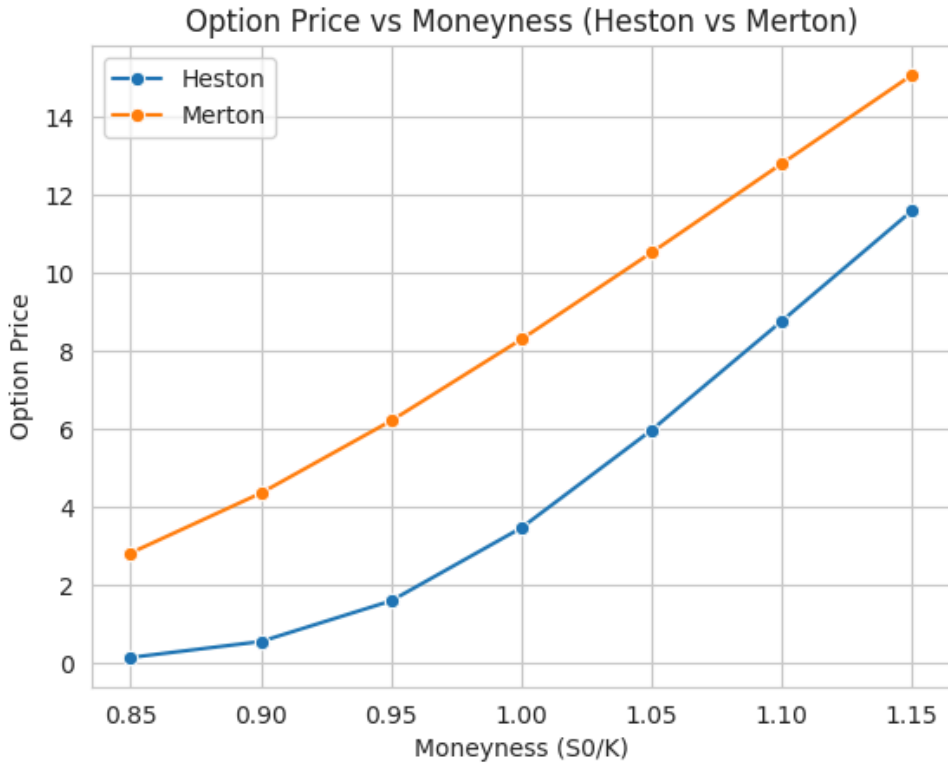


Fig: Option Price vs Moneyness for Heston & Merton model

Step 2:

Question 13: Early Exercise Right: American vs European Call

Results Comparison

Option Type	Price	Delta	Gamma
European Call (Q5, $\rho = -0.30$)	$2.86	0.5653	0.0204
American Call (Q13, $\rho = -0.30$)	$5.37	1.0208	0.0383

Unlike a European call which can only be exercised at expiry, an American call grants the holder the right to exercise at any point before or on the expiration date. This flexibility is theoretically more valuable, but for non-dividend-paying stocks, classical finance theory holds that it is never optimal to exercise early, because the holder forfeits the remaining time value of the option.

The American call is priced at $5.37 compared to the European call's $2.86, a gap of $2.51. This difference is larger than traditional theory would predict (which says they should be equal

for non-dividend-paying stocks). The divergence here arises primarily from the simulation methodology used (Longstaff-Schwartz): under stochastic volatility, the algorithm may identify early exercise as locally optimal in certain simulated paths where volatility has spiked and the intrinsic value is high.

The delta of 1.0208 (effectively 1.0) and the higher gamma also suggest the American call is behaving more like a deeply in-the-money option in some paths, an artefact of how early exercise boundaries are approximated under a stochastic volatility regime.

For investors in non-dividend-paying stocks, paying a premium for American-style optionality is generally unnecessary. The early exercise feature has negligible practical value in this setting. Traders should prefer the cheaper European call unless there are specific corporate events (such as a special dividend) that might make early exercise worthwhile.

Question 14:

Up and Call is a barrier option that only becomes active “knocks in”, if the stock price touches or exceeds $95 at least once before the option expires. If the stock never reaches $95, the option expires worthless regardless of where the stock ends up at maturity.

Starting from $80, the stock must rally by at least 18.75% to activate this option. That is a demanding condition over a 3-month window.

Why Is the Price So Low?

The barrier and strike are both set at $95, the same level. This means the option can only pay off if the stock both touches $95 (activating it) AND ends above $95 at expiry. Both conditions must hold simultaneously. Under the strong negative co-movement assumption ($\rho = -0.70$), when the stock rallies toward $95, market uncertainty simultaneously falls. This makes it harder to sustain momentum above the barrier, compressing the probability of a payoff.

Many simulated paths that end above $95 at maturity never actually touch $95 during the life of the option, these paths contribute nothing to the barrier option’s value, whereas they would contribute to the plain European call.

The up-and-in call at $0.01 is almost worthless under current assumptions. While it is extraordinarily cheap (compared to the plain call at $1.02), an investor would only profit in the narrow scenario where the stock rallies sharply to $95 and stays there or above. This is a highly speculative, high-conviction instrument.

The comparison highlights an important principle: barrier options trade at a steep discount to plain vanilla options because they carry conditional payoff risk. Investors should only use them when they have high conviction about a specific price trajectory, not just a directional view.

Results Comparison

Option Type	Strike	Barrier	Price
Plain European Call (K=$95)	$95.00	N/A	$1.02
Up-and-In Call (K=$95, B=$95)	$95.00	$95.00	$0.01

P.S: The plain European call price at K=$95 is computed from the same Q6 model parameters for a like-for-like comparison.

Question 15. We price a down-and-in put option with:

- Strike $K = 65$
- Barrier $B = 65$

using the same parameters as in Q8.

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Down-and-In Put: 15.64
European Put: 15.64
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In this case, the down-and-in put price is identical to the European put price. This is because the initial stock price ($S_0 = 50$) is already below the barrier level ($B = 65$). As a result, the barrier condition is immediately satisfied for all simulated paths, and the option is activated from the beginning.

Therefore, the down-and-in put behaves exactly like a standard European put in this setup. The presence of jumps increases the likelihood of hitting the barrier, which can make the down-and-in option more valuable compared to a pure diffusion setting.